

Day 03 – Transformers & Data Source Collection for SLM

Name: Moturi Lalitha Sowjanya

Team: 4

Week: 02

Date: 19-06-26

1) OBJECTIVE

To understand the concept of Transformers, their purpose, basic architecture, and the role of the Attention Mechanism in language models. Also, to continue collecting and organizing relevant resources, such as PDFs, research papers, FAQs, articles, public datasets, and websites, for the proposed Medical Information Assistant SLM.

2) RESEARCH TOPICS

2.1. Transformers

2.1.1. What are Transformers?

Transformers are a deep learning architecture used in modern language models to understand and process text. They use an **attention mechanism** to focus on the most important words in a sentence and understand the relationship between them, even if they are far apart.

Unlike older models that read text one word at a time, Transformers can process all the words in a sentence at the same time, making them faster and more accurate. They are the foundation of many modern AI models, such as **ChatGPT, Gemini, Llama, Claude, and Mistral**.

2.1.2. Why Transformers they introduced?

Transformers were introduced to overcome the limitations of older language models such as **RNNs (Recurrent Neural Network) and LSTMs (Long Short-Term Memory)**, which process text **one word at a time** and often struggle to remember information from earlier parts of long sentences.

Transformers solve this problem by processing all the words in a sentence at the same time and using an **attention mechanism** to focus on the most important words. This helps them understand long-range relationships, improves accuracy, and makes training faster and more efficient.

Example:

Sentence:

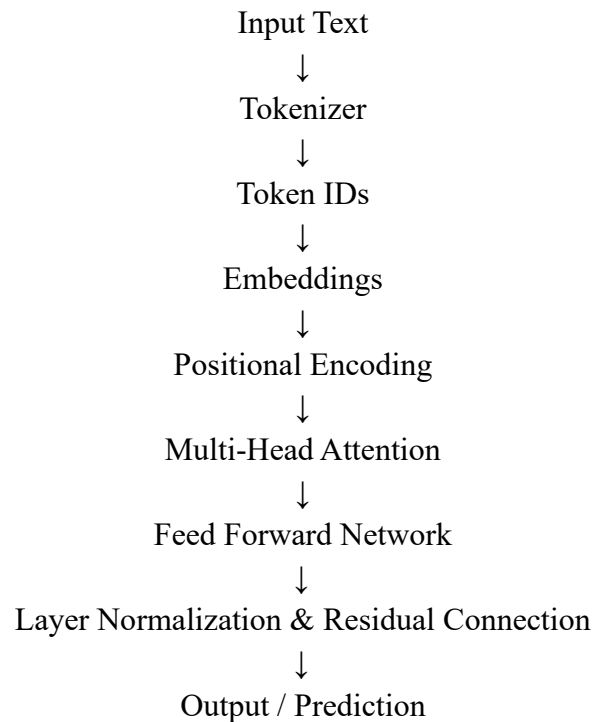
"The food that I ate yesterday at my grandmother's house was delicious."

- An RNN may forget the word **"food"** by the time it reaches **"delicious."**

- A Transformer can connect **"food"** and **"delicious"** even though they are far apart in the sentence.
-

2.1.3. Basic Architecture Overview

Transformer Workflow:



2.1.4. Components of a Transformer

1. Input Text:

The process begins with the text provided by the user.

Example:

"I love Artificial Intelligence."

2. Tokenizer:

The tokenizer breaks the input text into smaller units called tokens and converts them into numerical IDs that the model can process.

Example:

Input:

"I love AI"

↓

Tokens:

["I", "love", "AI"]

↓

Token IDs:

[101, 205, 876]

3. Embeddings:

Embeddings convert each token ID into a numerical vector that represents its meaning. Words with similar meanings have similar embeddings.

Example:

"I" → [0.12, -0.45, 0.81, ...]

"love" → [0.67, 0.15, -0.28, ...]

"AI" → [0.94, -0.10, 0.52, ...]

4. Positional Encoding:

Since Transformers process all words at the same time, they need positional encoding to understand the order of words in a sentence.

Example:

"Dog bites man"

"Man bites dog"

- Same words but different meaning
- Solution is Positional Encoding:

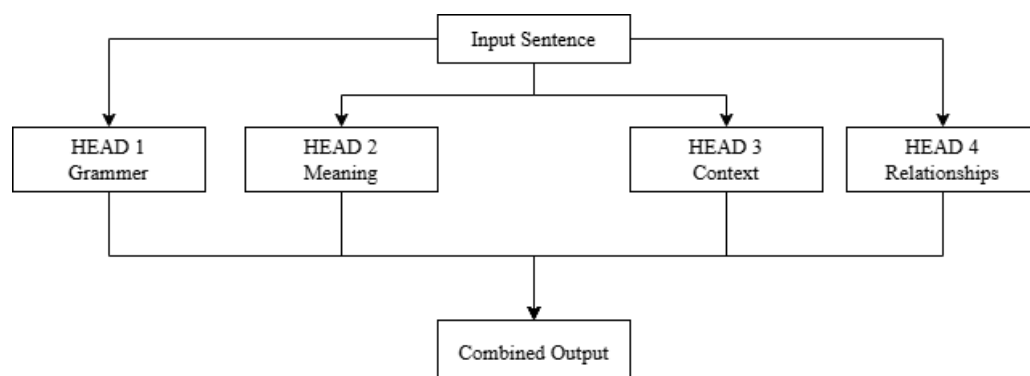
Word	Position
Dog	1
Bites	2
Man	3

5. Multi-Head Attention:

Multi-Head Attention is the core component of a Transformer. It helps the model focus on the most **important words and understand the relationships between them**.

Instead of using only one attention mechanism, the Transformer uses multiple attention heads. Each head learns different information, such as **grammar, meaning, context, or relationships**.

Workflow:



Example:

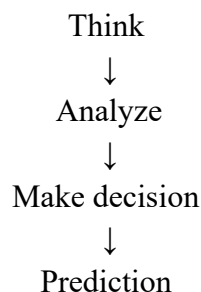
"The doctor prescribed medicine because the patient was sick."

Different attention heads focus on different information:

- **Head 1 (Grammar):** Understands the sentence structure.
- **Head 2 (Meaning):** Understands that the doctor prescribed the medicine.
- **Head 3 (Context):** Understands that the patient was sick.
- **Head 4 (Relationships):** Connects **patient** with **sick** and **doctor** with **prescribed medicine**.

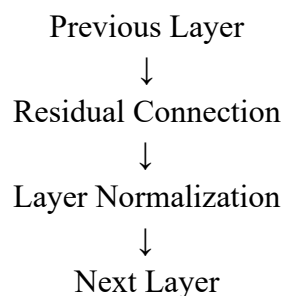
6. Feed Forward Network (FFN):

The Feed Forward Network processes the information received from the attention layer and learns more complex patterns before passing it to the next layer.

Workflow:**7. Residual Connection and Layer Normalization:**

Residual connections help **preserve important information** by passing it directly to the next layer, reducing the chance of losing information.

Layer normalization keeps the values balanced, making the model more stable and improving training.

Workflow:**8. Output/Prediction:**

After all the processing is complete, the Transformer generates the final output.

Example:

Input:

"I love AI"



Prediction:
"I love AI very much."

2.1.5. Role of Attention Mechanism.

The Attention Mechanism helps the Transformer focus on the most important words in a sentence. It understands how words are related to each other, even if they are far apart. This helps the model understand the meaning of the sentence more accurately.

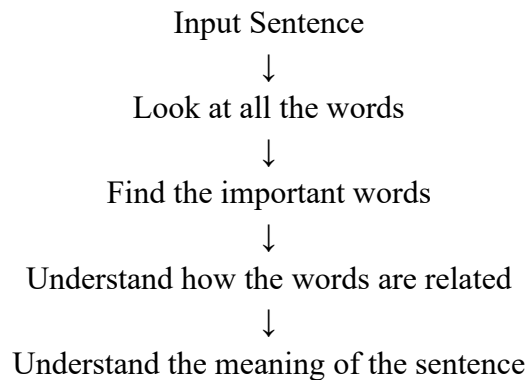
Example:

Sentence:

"The trophy doesn't fit in the suitcase because it is too large."

The attention mechanism understands that "it" refers to "the trophy", not the suitcase.

Workflow:



3) DATA COLLECTION FOR THE PROPOSED SLM

3.1. Collected Data Sources & Why These Data Sources are Useful

Data Source	Type	Description	Why is it Useful for the Proposed SLM?
DailyMed	Website	Contains information on diseases, symptoms, diagnosis, treatment, prevention, public health guidelines, and health awareness.	Helps the proposed Medical Information Assistant SLM provide reliable information about medicines, their uses, dosage, and safety precautions.

Data Source	Type	Description	Why is it Useful for the Proposed SLM?
Common Diseases	PDF	Contains information about common human diseases, including their causes, symptoms, diagnosis, prevention, treatment, infectious diseases, lifestyle diseases, cancer, sexually transmitted diseases, and drug abuse.	Provides comprehensive information about common diseases, their symptoms, causes, prevention, and treatment, helping the proposed Medical Information Assistant SLM answer a wide range of healthcare-related questions.
Infectious Diseases Handbook	PDF	Contains information about infectious diseases including their causes, symptoms, diagnosis, treatment, prevention, complications, and vaccination guidelines.	Helps the Medical Information Assistant SLM answer questions about infectious diseases, symptoms, diagnosis, treatments, prevention methods, and vaccines using detailed medical information.
Disease Prevention Handbook	PDF	Contains fact sheets on many infectious diseases, immunization guidelines, disease prevention, symptoms, treatment, and public health recommendations.	Helps the SLM answer questions about diseases, vaccines, prevention, symptoms, and infection control using reliable medical information.
Medical Handbook / Clinical Practice Guide	PDF	Contains information about common diseases, emergency conditions, medicines, diagnosis, treatment, prevention, prescription writing, and clinical management across multiple medical specialties.	Helps the proposed Medical Information Assistant SLM provide reliable information about diseases, symptoms, diagnosis, treatment, medicines, emergency care, and clinical guidelines across different areas of healthcare.

Data Source	Type	Description	Why is it Useful for the Proposed SLM?
WHO Standard Treatment Guidelines and Essential Medicines List	PDF	Contains standard treatment guidelines for common diseases and medical conditions, including diagnosis, treatment recommendations, emergency care, and an essential medicines list across multiple healthcare specialties.	Helps the proposed Medical Information Assistant SLM provide reliable information about diseases, diagnosis, treatment guidelines, emergency care, and essential medicines using trusted WHO recommendations

3.2. Links:

Website:

- <https://dailymed.nlm.nih.gov/dailymed/index.cfm>

PDFs:

- <https://share.google/e7D1d388oo9RWZDv1> - Common Disease
- <https://share.google/GYjg9BGKgtzeJuTkS> - Infectious Diseases Handbook
- <https://share.google/0wR0npqJW8YVvR83j> - Disease Prevention Handbook
- <https://share.google/uDsLhE1y5v8U26PMs> - Medical Handbook / Clinical Practice Guide
- <https://share.google/o2BznIZA3WCccZ0O> - WHO Standard Treatment Guidelines and Essential Medicines List